WORKED SOLUTION

STEP I 1989, Question 7

Pure • Curve Sketching and Optimisation

STEP I 1989, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s1-q7

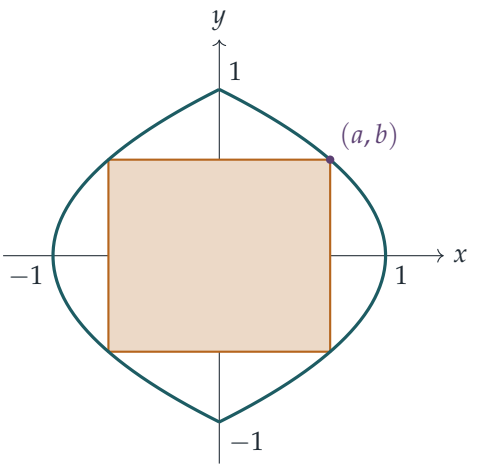
The key idea

The curve is symmetric in both axes, so the best rectangle should be centred on the origin with one corner on the curve in the first quadrant. That turns a two-variable search into maximising a function of one variable. For the general curve the same idea works once we solve for $|y|$ as a power of $1 - |x|$.

Solution

The sketch

Since $y^2 \geq 0$ we need $1 - |x| \geq 0$, so $-1 \leq x \leq 1$. For $x \geq 0$ the curve is $x = 1 - y^2$, a parabola opening to the left with vertex $(1, 0)$; for $x \leq 0$ it is its mirror image $x = y^2 - 1$. The two arcs meet at $(0, \pm 1)$, where they make corners (the slopes $\mathrm{d}y/\mathrm{d}x = \mp\frac{1}{2}$ and $\pm\frac{1}{2}$ do not match), and they cross the x-axis vertically at $(\pm 1, 0)$. The curve is a closed loop enclosing the region $|x| + y^2 \leq 1$, shaped like a rounded diamond.



The largest rectangle

Suppose a rectangle $[x_1, x_2] \times [y_1, y_2]$ lies inside the region $|x| + y^2 \leq 1$. Put $A = \max(|x_1|, |x_2|)$ and $B = \max(|y_1|, |y_2|)$. The corners are (x_i, y_j) with i, j chosen independently, so one corner has $|x| = A$ and $|y| = B$ at the same time. That corner lies in the region, so $A + B^2 \leq 1$. Then the rectangle $[-A, A] \times [-B, B]$ also lies in the region, because for $|x| \leq A$, $|y| \leq B$,

$$|x| + y^2 \leq A + B^2 \leq 1.$$

Its width $2A \geq x_2 - x_1$ and height $2B \geq y_2 - y_1$, so it is at least as big. Hence we may take the rectangle centred at the origin, with corners $(\pm a, \pm b)$, and pushing the corner out until it touches the curve, $a = 1 - b^2$ with $0 \leq b \leq 1$.

? **Ask yourself** Why is it enough to look at rectangles centred at the origin?

That is exactly the argument above: replacing each side by the larger of its two distances from the axis keeps the rectangle inside the curve (by symmetry) and never shrinks it.

The area is

$$S(b) = 4ab = 4b(1 - b^2) = 4b - 4b^3, \qquad S'(b) = 4 - 12b^2.$$

$S'(b) = 0$ at $b = 1/\sqrt{3}$, with $S' > 0$ before and $S' < 0$ after, and $S(0) = S(1) = 0$, so this is the maximum. There $a = 1 - \frac{1}{3} = \frac{2}{3}$ and

$$S_{\max} = 4 \cdot \frac{2}{3} \cdot \frac{1}{\sqrt{3}} = \frac{8}{3\sqrt{3}} = \frac{8}{\sqrt{27}},$$

as required.

The general curve

Now $y^{2m} = (1 - |x|)^n$. The left side is non-negative, and since n is odd, $(1 - |x|)^n$ has the same sign as $1 - |x|$. So again $|x| \leq 1$, and taking the (positive) $2m$-th root,

$$|y| = (1 - |x|)^k, \qquad k = \frac{n}{2m} > 0.$$

(If n were even, every x would be allowed and the curve would run off to infinity, so there would be no largest rectangle: this is why n must be odd.) The curve again is a closed loop through $(\pm 1, 0)$ and $(0, \pm 1)$, symmetric in both axes, enclosing $|y| \leq (1 - |x|)^k$. Since $(1 - |x|)^k$ decreases as $|x|$ increases, the same argument as before shows we may take a rectangle with corners $(\pm a, \pm b)$, $b = (1 - a)^k$, $0 \leq a \leq 1$. Its area is

$$S(a) = 4a(1 - a)^k.$$

Differentiating,

$$S'(a) = 4(1 - a)^{k-1}[(1 - a) - ka] = 4(1 - a)^{k-1}[1 - (1 + k)a],$$

which is positive for $0 < a < \frac{1}{1+k}$ and negative for $\frac{1}{1+k} < a < 1$. As $S(0) = S(1) = 0$, the maximum is at $a = \frac{1}{1+k}$, where $1 - a = \frac{k}{1+k}$:

$$S_{\max} = \frac{4}{1+k}\left(\frac{k}{1+k}\right)^k = \frac{4k^k}{(1+k)^{k+1}}.$$

With $k = \frac{n}{2m}$ and $1 + k = \frac{n+2m}{2m}$, the powers of $2m$ collect as $(2m)^{-k} \cdot (2m)^{k+1} = 2m$, so

$$S_{\max} = \frac{8m\, n^{n/(2m)}}{(n+2m)^{(n+2m)/(2m)}} = 8m\left(\frac{n^n}{(n+2m)^{n+2m}}\right)^{1/(2m)}.$$

Checks

- $m = n = 1$ gives $8\left(\frac{1}{27}\right)^{1/2} = 8/\sqrt{27} \approx 1.5396$, agreeing with the first part.
- A fine search over $a \in [0,1]$ for $(m,n) = (1,3), (2,1), (2,3), (3,5)$ matches the formula; for instance $(m,n) = (2,1)$ gives $16/5^{5/4} \approx 2.1400$. Random off-centre rectangles never beat it.
- As $m \to \infty$ with n fixed, $k \to 0$ and the curve fills out the square $|x| \le 1$, $|y| \le 1$; the formula tends to 4, the area of that square.

Common slips

- Forgetting the factor 4: the corner (a, b) gives a rectangle of width $2a$ and height $2b$.
- Ignoring why n odd matters: with n even there is no bounded curve.

Beyond STEP

Maximising $a(1-a)^k$ is a weighted AM–GM problem in disguise: writing $a \cdot \left(\frac{1-a}{k}\right)^k$ as a product of $1 + k$ factors with constant sum gives the maximum without calculus. The curves $|x|^p + |y|^q = 1$ are cousins of the unit balls of the ℓ^p norms, which appear in first-year analysis and linear algebra. For more on inequalities of this kind see Siklos, *Advanced Problems in Mathematics*, and for the analysis behind optimisation, Abbott, *Understanding Analysis*.

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